

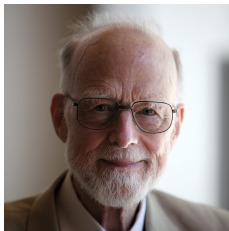
COMP2111 Week 8

Term 1, 2023

Hoare Logic

Sir Tony Hoare

- Pioneer in formal verification
- Invented: Quicksort,
- the null reference (called it his “billion dollar mistake”)
- CSP (formal specification language), and
- Hoare Logic



Summary

- $\mathcal{L}$: A simple imperative programming language
- Hoare triples (SYNTAX)
- Hoare logic (PROOF)
- Semantics for Hoare logic

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Imperative Programming

imperō

Definition

Imperative programming is where programs are described as a series of *statements* or commands to manipulate mutable *state* or cause externally observable *effects*.

States may take the form of a *mapping* from variable names to their values, or even a model of a CPU state with a memory model (for example, in an *assembly language*).

$\mathcal{L}$: A simple imperative programming language

Consider the vocabulary of basic arithmetic:

- Constant symbols: $0, 1, 2, \dots$
- Function symbols: $+, *, \dots$
- Predicate symbols: $<, \leq, \geq, |, \dots$

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- An **(arithmetic) expression** is a term over this vocabulary.
- A **boolean expression** is a predicate formula over this vocabulary.

The language $\mathcal{L}$

The language $\mathcal{L}$ is a simple imperative programming language made up of four statements:

Assignment: $x := e$

where x is a variable and e is an arithmetic expression.

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Conditional: if g then P else Q fi

where g is a boolean expression.

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Assignment: $x := e$

where x is a variable and e is an arithmetic expression.

Sequencing: $P; Q$

Conditional: if g then P else Q fi

where g is a boolean expression.

While: while g do P od

Factorial in $\mathcal{L}$

Example

```
 $i := 0;$   
 $m := 1;$   
while  $i < N$  do  
   $i := i + 1;$   
   $m := m * i$   
od
```

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Hoare Logic

We'll define a *Hoare Logic* for $\mathcal{L}$ to allow us to prove properties of our program.

We write a *Hoare triple* judgement as:

$$\{\varphi\} P \{\psi\}$$

Where φ and ψ are logical formulae about states, called *assertions*, and $P \in \mathcal{L}$. This triple states that if the program P terminates successfully from a starting state satisfying the *precondition* φ , then the final state will satisfy the *postcondition* ψ .

Hoare triple: Examples

Example

$$\{(x = 0)\} x := 1 \{(x = 1)\}$$

Hoare triple: Examples

Example

$$\{(x = 0)\} x := 1 \{(x = 1)\}$$
$$\{(x = 499)\} x := x + 1 \{(x = 500)\}$$

Hoare triple: Examples

Example

$$\{(x = 0)\} x := 1 \{(x = 1)\}$$

$$\{(x = 499)\} x := x + 1 \{(x = 500)\}$$

$$\{(x > 0)\} y := 0 - x \{(y < 0) \wedge (x \neq y)\}$$

Hoare triple: Factorial Examples

Example

```
 $\{N \geq 0\}$   
 $i := 0;$   
 $m := 1;$   
while  $i < N$  do  
   $i := i + 1;$   
   $m := m * i$   
od  
 $\{m = N!\}$ 
```

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We know what we want informally; how do we establish when a triple is valid?

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- Develop a semantics, OR

Hoare logic consists of one axiom and four inference rules for deriving Hoare triples.

Motivation

Question

We know what we want informally; how do we establish when a triple is valid?

- Develop a semantics, OR
- Derive the triple in a syntactic manner (i.e. Hoare proof)

Hoare logic consists of one axiom and four inference rules for deriving Hoare triples.

Assignment

$$\frac{}{\{\varphi[e/x]\} \textcolor{blue}{x} := \textcolor{blue}{e} \{\varphi\}} \quad (\text{assign})$$

Intuition:

If x has property φ *after* executing the assignment; then e must have property φ *before* executing the assignment

Assignment: Example

Example

$$\{(y = 0)\} x := y \{(x = 0)\}$$

Assignment: Example

Example

$$\{(y = 0)\} x := y \{(x = 0)\}$$
$$\{ \quad \quad \} x := y \{(x = y)\}$$

Assignment: Example

Example

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Assignment: Example

Example

$\{(y = 0)\} x := y \{(x = 0)\}$

$\{(y = y)\} x := y \{(x = y)\}$

$\{ \quad \quad \} x := 1 \{(x < 2)\}$

Assignment: Example

Example

$$\{(y = 0)\} x := y \{(x = 0)\}$$

$$\{(y = y)\} x := y \{(x = y)\}$$

$$\{(1 < 2)\} x := 1 \{(x < 2)\}$$

$$\{(y = 3)\} x := y \{(x > 2)\}$$

Assignment: Example

Example

$$\{(y = 0)\} x := y \{(x = 0)\}$$

$$\{(y = y)\} x := y \{(x = y)\}$$

$$\{(1 < 2)\} x := 1 \{(x < 2)\}$$

$$\{(y = 3)\} x := y \{(x > 2)\} \quad \textit{Problem!}$$

Sequence

$$\frac{\{\varphi\} P \{\psi\} \quad \{\psi\} Q \{\rho\}}{\{\varphi\} P; Q \{\rho\}} \quad (\text{seq})$$

Intuition:

If the postcondition of P matches the precondition of Q we can sequentially combine the two program fragments

Sequence: Example

Example

$$\frac{\{ \quad \} x := 0 \{ \quad \} \quad \{ \quad \} y := 0 \{ (x = y) \}}{\{ \quad \} x := 0; y := 0 \{ (x = y) \}} \quad (\text{seq})$$

Sequence: Example

Example

$$\frac{\{ \quad \} x := 0 \{ (x = 0) \} \quad \{ (x = 0) \} y := 0 \{ (x = y) \}}{\{ \quad \} x := 0; y := 0 \{ (x = y) \}} \quad (\text{seq})$$

Sequence: Example

Example

$$\frac{\{(0 = 0)\} x := 0 \{(x = 0)\} \quad \{(x = 0)\} y := 0 \{(x = y)\}}{\{(0 = 0)\} x := 0; y := 0 \{(x = y)\}} \quad (\text{seq})$$

Conditional

$$\frac{\{\varphi \wedge g\} P \{\psi\} \quad \{\varphi \wedge \neg g\} Q \{\psi\}}{\{\varphi\} \text{ if } g \text{ then } P \text{ else } Q \text{ fi } \{\psi\}} \quad (\text{if})$$

Intuition:

- When a conditional is executed, either P or Q will be executed.
- For the postcondition ψ to be established, *either* branch must terminate in a state satisfying ψ .

While

$$\frac{\{\varphi \wedge g\} P \{\varphi\}}{\{\varphi\} \text{ while } g \text{ do } P \text{ od } \{\varphi \wedge \neg g\}} \quad (\text{loop})$$

Intuition:

- φ is a **loop invariant**. It must be both a pre- and postcondition of P , so that sequences of P s can be run together.
- If the while loop terminates, g cannot hold.

Consequence

There is one more rule, called the *rule of consequence*, that we need to insert ordinary logical reasoning into our Hoare logic proofs:

$$\frac{\varphi' \rightarrow \varphi \quad \{\varphi\} P \{\psi\} \quad \psi \rightarrow \psi'}{\{\varphi'\} P \{\psi'\}} \quad (\text{cons})$$

Consequence

There is one more rule, called the *rule of consequence*, that we need to insert ordinary logical reasoning into our Hoare logic proofs:

$$\frac{\varphi' \rightarrow \varphi \quad \{\varphi\} P \{\psi\} \quad \psi \rightarrow \psi'}{\{\varphi'\} P \{\psi'\}} \quad (\text{cons})$$

Intuition:

- Adding assertions to the precondition makes it more likely the postcondition will be reached
- Removing assertions from the postcondition makes it more likely the postcondition will be reached
- If you can reach the postcondition initially, then you can reach it in the more likely scenario

Back to Assignment Example

Example

$\{(y = 3)\} x := y \{(x > 2)\}$ *Problem!*

Back to Assignment Example

Example

$\{(y = 3)\} x := y \{(x > 2)\}$ *Problem!*

$\{(y > 2)\} x := y \{(x > 2)\} (assign)$

Back to Assignment Example

Example

$\{(y = 3)\} x := y \{(x > 2)\}$ *Problem!*

$\{(y = 3)\} x := y \{(x > 2)\} (assign, cons)$
 $\{(y > 2)\} x := y \{(x > 2)\} (assign)$

Factorial Example

Let's verify the Factorial program using our Hoare rules:

$\{N \geq 0\}$

$i := 0;$
 $m := 1;$

while $i < N$ do

$i := i + 1;$

$m := m \times i$

od

$\{m = N!\}$

$$\frac{\{\varphi \wedge g\} P \{\psi\} \quad \{\varphi \wedge \neg g\} Q \{\psi\}}{\{\varphi\} \text{ if } g \text{ then } P \text{ else } Q \text{ fi } \{\psi\}}$$

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$i := 0;$
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od $\{m = i! \wedge N \geq 0 \wedge i = N\}$
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while $i < N$ do $\{m = i! \wedge N \geq 0 \wedge i < N\}$

$\{m \times (i + 1) = (i + 1)! \wedge N \geq 0\}$

$i := i + 1;$

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note: $(i + 1)! = i! \times (i + 1)$

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Factorial Example

Let's verify the Factorial program using our Hoare rules:

```

{N ≥ 0}
    i := 0;
    {1 = i! ∧ N ≥ 0} m := 1; {m = i! ∧ N ≥ 0}
    {m = i! ∧ N ≥ 0}
    while i < N do {m = i! ∧ N ≥ 0 ∧ i < N}
        {m × (i + 1) = (i + 1)! ∧ N ≥ 0}
        i := i + 1;
        {m × i = i! ∧ N ≥ 0}
        m := m × i
        {m = i! ∧ N ≥ 0}
    od {m = i! ∧ N ≥ 0 ∧ i = N}
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```

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$$\begin{array}{l}
 \{N \geq 0\} \\
 \quad i := 0; \{1 = i! \wedge N \geq 0\} \\
 \{1 = i! \wedge N \geq 0\} \quad m := 1; \{m = i! \wedge N \geq 0\} \\
 \{m = i! \wedge N \geq 0\} \\
 \text{while } i < N \text{ do } \{m = i! \wedge N \geq 0 \wedge i < N\} \\
 \quad \{m \times (i + 1) = (i + 1)! \wedge N \geq 0\} \\
 \quad i := i + 1; \\
 \quad \{m \times i = i! \wedge N \geq 0\} \\
 \quad m := m \times i \\
 \quad \{m = i! \wedge N \geq 0\} \\
 \text{od } \{m = i! \wedge N \geq 0 \wedge i = N\} \\
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 \end{array}$$

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note: $(i + 1)! = i! \times (i + 1)$

Factorial Example

Let's verify the Factorial program using our Hoare rules:

```

{N ≥ 0}
{1 = 0! ∧ N ≥ 0} i := 0; {1 = i! ∧ N ≥ 0}
{1 = i! ∧ N ≥ 0} m := 1; {m = i! ∧ N ≥ 0}
{m = i! ∧ N ≥ 0}
while i < N do {m = i! ∧ N ≥ 0 ∧ i < N}
  {m × (i + 1) = (i + 1)! ∧ N ≥ 0}
  i := i + 1;
  {m × i = i! ∧ N ≥ 0}
  m := m × i
  {m = i! ∧ N ≥ 0}
od {m = i! ∧ N ≥ 0 ∧ i = N}
{m = N!}
    
```

$$\frac{\{\varphi \wedge g\} P \{\psi\} \quad \{\varphi \wedge \neg g\} Q \{\psi\}}{\{\varphi\} \text{ if } g \text{ then } P \text{ else } Q \text{ fi } \{\psi\}}$$

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note: $(i + 1)! = i! \times (i + 1)$

Practice Exercise

Example

```
 $m := 1;$   
 $n := 1;$   
 $i := 1;$   
while  $i < N$  do  
   $t := m;$   
   $m := n;$   
   $n := m + t;$   
   $i := i + 1$   
od
```

Practice Exercise

Example

```
 $m := 1;$   
 $n := 1;$   
 $i := 1;$   
while  $i < N$  do  
   $t := m;$   
   $m := n;$   
   $n := m + t;$   
   $i := i + 1$   
od
```

- What does this $\mathcal{L}$ program P compute?
- What is a valid Hoare triple $\{\varphi\}P\{\psi\}$ of this program?
- Prove using the inference rules and consequence axiom that this Hoare triple is valid.

Summary

- $\mathcal{L}$: A simple imperative programming language
- Hoare triples (SYNTAX)
- Hoare logic (PROOF)
- Semantics for Hoare logic

Semantics

Nope. That's a topic for another lecture.